

MUAN MEURER

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EDUCATION

Carnegie Mellon University - Pittsburgh, PA

August 2025 - December 2026 (expected)

Master of Science in Mechanical Engineering

Relevant Coursework: *Robot Dynamics, Applied FEA, Space Robotics, Mobile Robots, Electromechanical Systems Design*

California State University, Los Angeles – Los Angeles, CA

August 2017 - May 2022

Bachelor of Science in Physics | The Early Entrance Program | GPA 3.86/4.00

PROJECTS

NASA Lunabotics - CMU | *Project Lead & Mechanical Engineer*

September 2025 - Present

- Led 73-person team to develop end-to-end design of a lunar rover for autonomous berm construction, spanning chassis, drivetrain, excavation, and avionics interfaces.
- Authored system-level requirements, ICDs, and risk/verification plans following SRR/PDR/CDR review structure.
- Designed and analyzed load-bearing structures for excavation and mobility in abrasive lunar regolith, accounting for traction limits and duty-cycle loading.
- Performed FEA-driven design iteration (ANSYS) to test stress, deflection, and safety margins before fabrication.
- Drove integration with electrical and software teams, resolving mechanical-avionics constraints around sensors, harnessing, and serviceability.

Automated Coil Winder - Westinghouse & CMU | *Mechatronics Engineer*

January 2026 - Present

- Designed and built an automated variable-diameter helical coil winder for Westinghouse's eVinci™ microreactor team to prototype high-fidelity LVDT sensor coils.
- Developed a lathe-style mechatronic winding system with spindle rotation, synchronized linear wire traversal, passive tensioning, and interchangeable mandrels for repeatable coil fabrication.
- Collaborated weekly with Westinghouse engineers to document design decisions, validate test results, and deliver a functional prototype capable of producing inductance coils with less than 3% inductance variability.

Moonranger - CMU | *Mechanical, Thermal, and Battery Sim Engineer*

August 2025 - Present

- Built 1500+ node Thermal Desktop models to evaluate lunar rover thermal survivability across transit heat cycles.
- Modeled conduction, radiation, and internal dissipation to assess component stress under worst-case hot/cold cases.
- Coupled thermal, mechanical, and battery models to ensure power system viability over a 14-day lunar surface operation window.
- Developed Python simulations to quantify terrain-driven power losses (~20%) and inform system energy budgeting.
- Supported design trades by translating thermal results into mechanical design constraints and operational limits.

The Nerenberg Lab – CSULA | *Biophysics Research Student*

November 2021 - May 2022

- Ran large HPC simulations (10M+ timesteps) to simulate numerically stable/convergent protein folding.
- Designed and validated simulation pipelines, stress-testing assumptions, and documented failure modes.
- Authored a 60-page technical thesis detailing methodology robustness, uncertainty sources, and reproducibility.

PROFESSIONAL EXPERIENCE

V3 Electric | *Solar Sales Consultant*

July 2024–July 2025

- Developed working knowledge of solar energy systems, translating technical specs into customer-facing designs.
- Closed \$300K/month in revenue, demonstrating disciplined execution and outstanding stakeholder communication.

LEADERSHIP

The Early Entrance Program Club | *President*

May 2021 - May 2022

- Led a 20-member executive council, managed a \$10K+ budget, and coordinated programming for 100 participants.

SKILLS

Mechanical & Hardware: SolidWorks, Fusion 360, GD&T, DFM, structural design, drivetrain systems, rapid prototyping

Analysis & Simulation: ANSYS Mechanical/Workbench, FEA, Thermal Desktop, safety margins, trade studies

Programming & Controls: Python, Path Planning Algorithms, MATLAB, C++, Arduino, PID motor control